

Least Squares ● Intermediate

The formal rule that picks exactly one line out of the infinite candidates.

ORDINARY LEAST SQUARES (OLS)

Choose the line that minimizes the total squared residual error across every data point in the training set.

FORMULA

$$\text{SSE} = \sum (y_i - \hat{y}_i)^2$$

y_i is the actual value of observation i , $\hat{y}_i$ is the model's prediction for observation i , the subtraction gives the residual, squaring removes cancellation, and Σ (sigma) sums this over every observation.

A complete tiny numerical example

Suppose a candidate line predicts: for $x = [1, 2, 3]$, $\hat{y} = [2, 4, 6]$; actual $y = [3, 3, 7]$.

x	Actual y	Predicted $\hat{y}$	Residual	Squared residual
1	3	2	1	1
2	3	4	-1	1
3	7	6	1	1
SSE				3

Now compare a second candidate line, $\hat{y} = [3, 4, 5]$:

x	Actual y	Predicted $\hat{y}$	Residual	Squared residual
1	3	3	0	0
2	3	4	-1	1
3	7	5	2	4
SSE				5

REMEMBER THIS

The first candidate line ($SSE = 3$) fits better than the second ($SSE = 5$) — even though the second line predicts one point *exactly*. OLS cares about the total, not any single point.

The Mathematics Behind the Line ● Intermediate

Deriving b_0 and b_1 by hand — the formula stops being magic once you calculate it once.



SIMPLE LINEAR REGRESSION FORMULAS

$$b_1 = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / \Sigma[(x_i - \bar{x})^2]$$

$$b_0 = \bar{y} - b_1\bar{x}$$

$\bar{x}$ ("x-bar") is the mean of x ; $\bar{y}$ is the mean of y . $(x_i - \bar{x})$ is how far a single x -value deviates from the average x . The numerator measures how x and y move *together* (their covariance); the denominator measures how much x varies on its own. The ratio is, intuitively, "how much does y move per unit that x moves," which is exactly what a slope means.

Manual walkthrough with a tiny dataset

Data: $x = [1, 2, 3]$, $y = [2, 4, 5]$. Mean of $x = 2$, mean of $y = 3.667$.

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
1	2	-1	-1.667	1.667	1
2	4	0	0.333	0	0
3	5	1	1.333	1.333	1
Sum				3.0	2.0

$b_1 = 3.0 / 2.0 = 1.5$. $b_0 = 3.667 - (1.5 \times 2) = 0.667$. Final equation: $\hat{y} = 0.667 + 1.5x$.

INTERVIEW TIP

If asked "why does this formula produce the minimum-SSE line," the honest short answer is: it's the result of taking the derivative of the SSE function with respect to b_0 and b_1 , setting both to zero, and solving the resulting two equations. You don't need to reproduce that derivation from scratch in most interviews — but say this out loud instead of pretending the formula fell from the sky.

A Full Worked Example Intermediate

Start to finish, by hand: Advertising Spend predicting Sales.

Advertising Spend	Sales
1	2
2	4
3	5
4	7
5	8

Step 1–2: $\bar{x} = 3$, $\bar{y} = 5.2$

x	y	$x - \bar{x}$	$y - \bar{y}$	Product	$(x - \bar{x})^2$
1	2	-2	-3.2	6.4	4
2	4	-1	-1.2	1.2	1
3	5	0	-0.2	0.0	0
4	7	1	1.8	1.8	1
5	8	2	2.8	5.6	4
Sum				15.0	10

Steps 6–8: $b_1 = 15/10 = 1.5$. $b_0 = 5.2 - 1.5 \times 3 = 0.7$. Regression equation: $\hat{y} = 0.7 + 1.5x$.

Step 9 — a prediction: at spend = 6, $\hat{y} = 0.7 + 1.5 \times 6 = 9.7$.

Steps 10–11 — residuals and SSE:

x	Actual y	$\hat{y}$	Residual	Squared
1	2	2.2	-0.2	0.04
2	4	3.7	0.3	0.09
3	5	5.2	-0.2	0.04
4	7	6.7	0.3	0.09
5	8	8.2	-0.2	0.04
SSE				0.30

REMEMBER THIS

This exact dataset drives the interactive playground in Chapter 4 — go drag those points and watch these same numbers change live.